

Low-Cost Geometric Directions in Frozen Speech Models: Where a Training-Free Deepfake-Hardness Signal Helps—and Where It Does Not

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Abstract

Deepfake-speech detectors are inconsistent in two systematic ways: against a fixed detector some synthesis systems evade detection far more often than others, and a detector trained on one recording domain often fails on another. Working entirely at small scale—one GPU, a frozen $\sim 95\text{M}$ -parameter encoder, no detector retraining—we show that much of this inconsistency is legible in the *geometry* of the frozen representation. Inside the encoder there is a single natural-versus-synthetic direction, and how far a synthesis system’s utterances *spread* along it predicts how hard that system is to detect—the only quantity, out of 83 we tested, to survive a global false-discovery-rate correction. Two further results follow. The direction is not universal: it rotates across datasets to near-orthogonality, the geometric reason detectors fail to transfer. And it is a training-free corrective signal conditional on headroom: it cuts a weak detector’s error by $\approx 21\%$ while leaving an already-strong one unchanged. Every quantitative claim is independently recomputed.

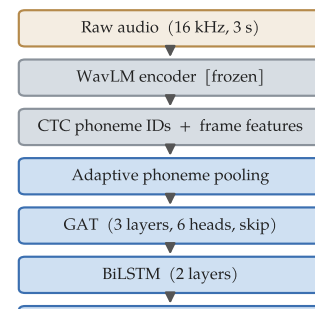
1 Introduction

Synthetic speech is now realistic enough that its detection is a security necessity, and the standard defence is a classifier that flags synthetic utterances. Yet detectors are unreliable in two ways that aggregate accuracy hides: some synthesis systems evade a fixed detector far more often than others (Wang et al., 2020; Müller et al., 2024a), and a detector tuned on one recording domain often fails on another (Müller et al., 2022). Neither had a clean explanation.

We provide one, at small scale. Modern detectors sit on top of a large frozen self-supervised (SSL) speech encoder (Chen et al., 2022; Tak et al., 2022; Wang & Yamagishi, 2022); we study that frozen representation before any decision boundary is drawn. Our central finding is a **natural-versus-synthetic axis**, a single direction from natural to synthetic speech: a system whose utterances stay tight along it is detected easily, one whose utterances spread along it is hard (Fig. 2). That spread, sd_{along} , is our predictor. Three results follow. (i) *The law*: sd_{along} predicts per-system hardness and is the only one of 83 tested quantities to survive global multiple-testing correction (Sec. 3.2). (ii) *The axis rotates*: each dataset has its own axis, so a transferred axis is uninformative—the geometric reason transfer fails. (iii) *A training-free corrective signal*: fusing the axis into a detector’s score needs no detector training and no difficulty labels, and helps only when the detector has headroom (Sec. 4). Every reported number was independently recomputed from cached artifacts (Sec. 3.3).

2 Setup

Detector. Our primary detector (WavLM-GAT) is the phoneme-level graph-attention model of Zhang et al. (2025): a frozen WavLM encoder feeds adaptive phoneme pooling, a 3-layer 6-head graph-attention network (Veličković et al., 2018), a BiLSTM, and a linear



41 head (Fig. 1). The encoder is frozen ($\sim 95\text{M}$ parameters;
 42 WavLM-base, 94.7M); only the $\sim 13\text{M}$ -parameter head
 43 trains. For architecture generality we also use AASIST
 44 (Jung et al., 2022), a spectro-temporal detector *not* built
 45 on WavLM.

46 **Corpora and hardness.** The law is established on
 47 MLAAD (Müller et al., 2024b) (61 text-to-speech systems,
 48 ≥ 8 utterances each); the lever and rotation check add
 49 ASVspoof 2019 LA (Wang et al., 2020). Per-system hard-
 50 ness is $1 - \text{AUC}$ of the detector’s scores for that system
 51 against a shared bona-fide pool. Class overlap is exactly
 52 what lowers AUC, so a geometric “spread across the
 53 boundary” should map onto hardness.

54 **Objective hardness.** Per-system hardness is a stable tar-
 55 get, not a quirk of one run. Three independently seeded
 56 WavLM-GAT detectors on MLAAD agree on the per-
 57 system hardness ranking at Spearman $\rho = 0.89\text{--}0.97$ (pair-
 58 wise, 61 systems); on ASVspoof 2019 LA the three seeds converge on essentially the same
 59 ranking ($\rho = 0.94\text{--}1.00$, zero difficulty-category flips). Hardness is a property of the (archi-
 60 tecture, recording domain) pair, worth predicting, not of a random seed.

61 **Compute.** Everything runs on one GPU; the largest single training run is $\approx 10^{16}$ FLOPs
 62 (well under the 10^{20} budget and 3B-parameter cap).

63 3 Experiments

64 3.1 The axis and its spread

65 **Frozen embeddings.** For each utterance (16 kHz, 3s)
 66 we take the frozen WavLM layer-12 hidden states and
 67 mean-pool over time, one 768-dimensional vector per
 68 clip. We read geometry from layer 12 because information
 69 in SSL encoders stratifies from acoustic detail in early
 70 layers to linguistic content in deep layers (Pasad et al.,
 71 2021; 2024): the natural-versus-synthetic distinction is
 72 linguistic, so the deepest layer is where it should—and
 73 empirically does—live. Each coordinate is standardized
 74 by the training bona-fide mean and standard deviation,
 75 so none dominates through raw scale.

76 **The axis.** Per corpus we estimate the direction as the stan-
 77 dardized centroid difference $w = \text{centroid}(\text{synthetic}) -$
 78 $\text{centroid}(\text{bona})$ —a two-group mean difference, not a
 79 trained classifier: it uses the bona/synthetic *group* labels but no fitted decision bound-
 80 ary and no per-system difficulty labels. We estimate w corpus-internally (the direction
 81 differs across corpora, Sec. 3.2) and **leave-one-system-out** (LOSO): when scoring a system
 82 we rebuild w from the bona pool plus every *other* system’s synthetic clips, so a held-out
 83 system never informs the axis used to judge it—without LOSO a system would help define
 84 the ruler it is graded on. For each system we record sd_{along} , the standard deviation of its
 85 projections onto w ; the mechanism (Fig. 2) predicts this second moment beats mean position.

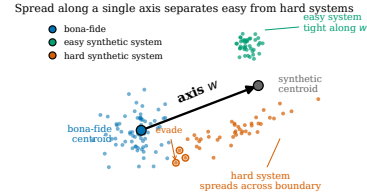
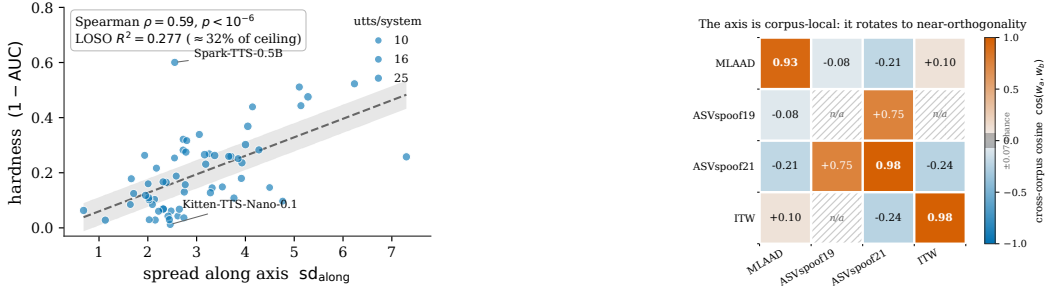


Figure 2: An easy system stays tight along w ; a hard system spreads across the boundary, so some utterances land in the bona-fide region and evade. Spread, not mean position, is the predictor.

86 3.2 Results: the law and its rotation

87 **A natural-synthetic axis predicts detection hardness.** Across 61 MLAAD systems, hardness
 88 rises with sd_{along} (Fig. 3a; Spearman $\rho = 0.60$, $p < 10^{-6}$; LOSO $R^2 = 0.277$, permutation
 89 $p = 2.5 \times 10^{-4}$; bootstrap ρ CI $[0.41, 0.73]$). Systems that straddle the boundary produce
 90 utterances that look genuine to the encoder, and those evade. Mean position along the



(a) The hardness law (MLAAD, 61 systems).

(b) The axis is corpus-local.

Figure 3: **(a)** Spread along the corpus-internal axis predicts MLAAD hardness. **(b)** Cosine between per-corpus axes: within-corpus split-half reliability on the diagonal (0.93–0.98); cross-corpus cosines near zero or negative, inside/at the ± 0.07 chance band—a transferred axis is uninformative.

axis—the naive “closer to bona is harder” guess—does *not* survive; the second-moment quantity does. Of the 83 quantities tested across the study, sd_{along} is the *only* one to survive a global Benjamini–Hochberg false-discovery-rate correction (Benjamini & Hochberg, 1995). It survives leave-one-system jackknifing (ρ range 0.58–0.63) and strengthens as measurement noise falls: $\rho = 0.60$ at ≥ 8 utterances/system ($n=61$), 0.64 at ≥ 12 ($n=51$), and 0.69 at ≥ 16 ($n=28$)—the attenuation signature of a real effect.

Directional, not circular, and not WavLM-specific. A centroid axis could resemble the detector’s own logit direction, but does not: substituting the representation’s top discriminative directions with class-blind values degrades detection ($\Delta\text{AUC} = -0.032$), while the same operation on the equal-magnitude orthogonal residual does not ($+0.004$), across all three seeds. The law also replicates on a different architecture: AASIST shares none of WavLM’s geometry, so its per-system hardness is an independent target (correlating only ≈ 0.55 with WavLM-GAT’s), yet the WavLM-derived sd_{along} still ranks it ($\rho = +0.35, p = 0.006$). The rank-order form replicates; the R^2 regression form does not on AASIST ($r^2 \approx 0$), and we report both.

The axis rotates across datasets. A predictor fit on one corpus fails on another because its direction is not the same direction elsewhere. Each corpus’s internal axis is estimated almost noiselessly—split-half reliability 0.93–0.98 (Fig. 3b, diagonal)—so two corpora that shared a natural–synthetic direction would show a large positive cross-axis cosine. They do not: MLAAD and ASVspoof-2019 have cosine ≈ -0.08 , essentially orthogonal, at the 768-dimensional chance floor ($|\mathbb{E}[\cos]| \approx 0.03$, 95th pct ≈ 0.07 ; Vershynin, 2018). We are careful about sign: at this magnitude the pair is best read as *unrelated*, not reliably reversed; the sign turns reliably negative only for larger-magnitude pairs (MLAAD–ASVspoof-2021 ≈ -0.21 , ITW–ASVspoof-2021 ≈ -0.24), several standard deviations beyond chance. Either way a transferred axis carries no usable information, so an axis-based diagnostic must be re-estimated on the target domain; part of the rotation may reflect recording-channel nuisance (Chen et al., 2021), but the lesson is unchanged.

3.3 Verification

A red-team pass recomputed every number from cached artifacts, returning 8 verified, 6 mislabeled, 5 misleading, and 2 contradicted claims, each corrected or removed: a cross-corpus cosine reported as $+0.36$ was in fact negative; an R^2 of 0.316 attributed to sd_{along} was a three-feature model, the single-feature value being 0.277; a “residual causal effect” disagreed in sign across seeds and was dropped. The multiplicity accounting is the global Benjamini–Hochberg procedure over all 83 recorded tests (Benjamini & Hochberg, 1995), with permutation p -values throughout (Ojala & Garriga, 2010), and $sd_{\text{along}} \rightarrow \text{hardness}$ is the sole survivor.

The law is not an MLAAD channel artifact: partialling the layer-0 velocity and energy channel proxies out of both sd_{along} and hardness leaves the association essentially undiminished

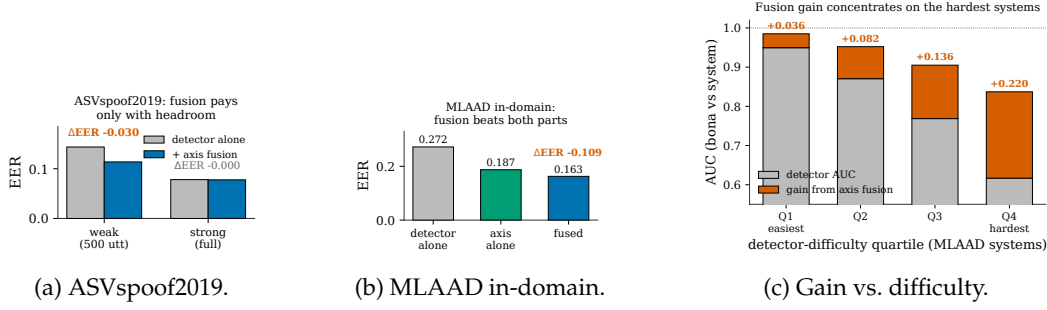


Figure 4: **(a)** The same recipe cuts a weak ASVspoof2019 detector’s EER (0.144 $\rightarrow$ 0.114) but leaves a strong, near-ceiling one unchanged. **(b)** In-domain on MLAAD, fusion (0.163) beats detector-alone (0.272) and axis-alone (0.187). **(c)** Fusion’s AUC gain rises from +0.036 (easiest MLAAD quartile) to +0.220 (hardest).

(partial Spearman $\rho = +0.58$, permutation $p = 1 \times 10^{-4}$; the proxies are non-significant in a joint regression), removing the most direct “it is just the microphone” explanation. The target is also reliable: hardness’s split-half correlation is 0.76, which the Spearman–Brown prophecy formula (Spearman, 1904) corrects to a full-length reliability of 0.86—the R^2 ceiling. So sd_{along} ’s $R^2 = 0.277$ captures $\approx 32\%$ of the *explainable* variance; 0.76 and 0.86 are the same reliability before and after the correction, not a discrepancy.

4 Axis fusion improves performance under detector headroom

Because the axis identifies *which* systems a detector will struggle with, its projection fuses into the score as $s_{\text{fused}} = z(\text{logit}) + \lambda z(\text{axis})$, needing no detector training and no difficulty labels. To avoid leakage, the z -standardization statistics and the weight λ are fit on each system-disjoint fold’s *training* partition only, never on the held-out evaluation utterances. Whether this helps depends entirely on headroom.

Headroom decides. A deliberately weak WavLM-GAT trained on 500 utterances drops from EER 0.144 to 0.114 under the unsupervised axis ($\Delta\text{EER} = -0.030$, 95% CI $[-0.046, -0.013]$, $p \approx 0$), improving all five attack-disjoint folds—a $\approx 21\%$ reduction (Fig. 4a). The identical procedure on a strong, near-ceiling WavLM-GAT (EER 0.078) produces no change beyond noise ($\Delta\text{EER} = -0.0004$ to $+0.004$, $p \approx 0.56$ –0.80). In-domain on MLAAD, likewise far from ceiling, the same centroid fusion improves EER from 0.272 to 0.163 (axis alone 0.187), negative across all folds and seeds (Fig. 4b).

Why the axis works. The gain lands on the hardest systems. By MLAAD detector-difficulty quartile, fusion’s AUC gain grows monotonically from +0.036 (easiest) to +0.220 (hardest) (Fig. 4c)—the law turned into performance. Hard systems sit close to bona-fide speech with high spread along a direction the detector under-weights, so re-injecting it recovers exactly them; a strong detector has already learned the direction and gains nothing—the headroom condition follows from the mechanism.

5 Limitations and conclusion

The law is text-to-speech-specific: on voice conversion all predictors give $\rho \approx 0$, as converted speech inherits its content, and spread, from source speech. Geometry uses one backbone (WavLM; wav2vec 2.0 (Baevski et al., 2020) and HuBERT (Hsu et al., 2021) untested), and the causal contrasts rest on three seeds—trustworthy in sign, not in power. Cross-dataset *prospective* prediction is unconfirmed: a pre-registered, externally timestamped test on held-out ASVspoof-2021-DF failed honestly ($\rho = +0.33$, $p = 0.135$, $n = 13$), reported not omitted. Within these bounds, a low-cost, frozen, training-free direction genuinely helps understand and partially correct detectors—for weak detectors within a dataset, not across datasets or for strong ones.

Reproducibility statement

All results use a single-GPU pipeline: a frozen $\sim 95\text{M}$ -parameter encoder (WavLM-base, 94.7M) and a $\sim 13\text{M}$ head ($\ll 3\text{B}$ parameters), largest run $\approx 10^{16}$ FLOPs. Every number re-generates from a committed script over cached embeddings, released with logs, checkpoints, and the audit.

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